

Bd Genetics Analysis Plan

Cannon Mallory

January 15, 2026

Contents

1	Helpful Papers	4
2	Introduction and preamble	5
3	Available Sequence Data	5
4	NGS Analysis	6
4.1	General NGS Analysis Notes	6
4.2	General steps	6
4.2.1	START: Pre-processed Sequencing Reads	6
4.2.2	QC:	6
4.3	Quality Filter (if needed)	6
4.3.1	Trim Low-quality ends:	7
4.4	Genome Size Estimation, Read Quality Estimate:	11
4.5	Polishing Genome Assembly (PacBio or Nanopor primarily):	19
4.6	Duplicate Marking and Indexing:	21
4.7	Alignment statistics	24
4.8	Concensus Sequence:	25
4.9	Final Assembly Assesement: (question: Is this step necessary for a genome resequencing and build from a reference genome? Theoretically this should be enough so long as our sequences had relatively good alignment with the reference genome. POnder.)	26
4.10	Choices	27
4.10.1	Option 1: Genome Annotation and partial genome phyloganies/variant analysis:	28
4.10.2	Variant (SNP and INDEL) Calling:	28
4.10.3	Option 2: Whole Genome comparative analysis and phylogenetic trees or PCAs.	29
5	PacBio Analysis	30
5.1	Generaly pacbio analysis notes	30
5.2	Haplotype level assembly tools for pacbio sequencing:	30
6	Population and intra population Phylogenetics	31
6.1	Phylogenetic Fundamentals:	31
6.2	Population phylogenetic fundamentals:	31
6.3	Phylogenetic tools:	31
7	Imani Sequences What I have done (in rough order with dates)	32
7.1	Genious Stuff	32
7.2	Bowtie alignment 1	32
7.3	Illumina Sequencing Analysis Real:	32

1 Helpful Papers

Before reading any part of this document I highly recommend reading the two great works of Ziheng Yang. His 2006 textbook of the title Computational Molecular Evolution and his 2014 update titled Molecular Evolution: A statistical Approach. These works are essential for considering and thinking about evolution and have set the standard in the field. If you are considering doing any amount of phylogenetics especially I would recommend the works of Ziheng Yang. Also refer to Sishuo Wang's github that has solutions to many of the questions and problems presented by Ziheng Yang in his books. The github can be found here: [`https://github.com/sishuowang/Solutions`](https://github.com/sishuowang/Solutions)_{*Manual_CME2006_MESA2014*}

Here are some papers that I found very helpful in the processes of coming up with this plan. I will attempt to cite which papers did what where but that may not be completely accurate during the main body of the text. Feel safe in knowing that these specific papers are the primary sources for all the information that I have compiled though!

- A beginner's guide to assembling a draft genome and analyzing structural variants with long-read sequencing technologies. [**kimBeginnersGuideAssembling2022**]
- Complex history of amphibian killing fungus reveals new insights into pathogen genotype and virulence. [**rosenblumComplexHistoryAmphibiankilling2013**]
- Chromosomal copy number variation in the amphibian killing fungus *Batrachochytrium dendrobatidis*. [**muletz-wolzDiverseGenotypesAmphibiankilling2019**]
- Hybrid Assembly of the Chytrid Genome - Insights into the Evolution and Outbreak of a Major Amphibian Pathogen
- Diploidy within a Haploid Genus of Entomopathogenic Fungi. [**nielsenDiploidyHaploidGenus2021**]
- Cryptic diversity of a widespread global pathogen reveals expanded threats to amphibian conservation. [**byrneCrypticDiversityWidespread2019**]
- Genomic Correlates of Virulence Attenuation in the Deadly Amphibian Chytrid Fungus, *Batrachochytrium dendrobatidis* [**refsniderGenomicCorrelatesVirulence2015**]
- Chapter 8 - Decoding the blueprint: an overview of genome sequencing, assembly, and annotation [**DecodingBlueprintOverview2025**]
- Genomic Perspectives on the Pathogenic Chytrid Fungus and Persisting Amphibian Hosts [**byrneGenomicPerspectivesPathogenic2020**]

2 Introduction and preamble

There are a couple different approaches to using sequence data for my applications. For this work I have a few related but ultimately independent goals:

1. Create phylogenies with older NGS sequence data
 - Mitochondrial
 - Genomic DNA
2. Create phylogenies with new PacBio long read data
 - Mitochondrial
 - Genomic DNA
3. Create CCNV Estimates of existing NGS sequence data
4. Create CCNV Estimates of new PacBio long read data

3 Available Sequence Data

1. **Published NGS Sequence Data:** There are approximately 40 published Bd NGS sequences from around the world. These data were all short read (150PE to 300PE) Illumina sequences.
2. **Published PacBio HiFi sequence:** There is a single pacbio hifi sequence published. I will generally reference this sequence by the name **near_complete** This sequence has good mitochondrial data!
3. **Imani Sequence:** Imani sequenced about 27 (double check number) bd isolates from across the California Sierra Nevada. These isolates were sequenced on an Illumina Sequencer PE300. I think they were NovaSeq but not 100% sure, I do know that they were sequenced on one of the newer Illumina machines! As far as I am aware these data have not been published in any way.
4. **New Sequence:** I hope to sequence some number of new sequences! Number TBD but if I am lucky these will be done on a PacBio hifi sequencer!

4 NGS Analysis

4.1 General NGS Analysis Notes

Creating a de-novo assembly from NGS sequence data is generally not super successful. It is most certainly possible however the results are generally less than desirable. When used for whole genome assembly NGS data can miss large structural variations in the genomes and might miss novel genes. It also struggles with highly repetitive regions. NGS data will, however, accurately capture SNPS at a reasonably high fidelity. The benefit of NGS data is predominantly the price.

The Good news, however, is we already have a published PacBio reference genome! If we already have a good reference genome then aligning illumina sequence data especially paired end sequences can be EXTREMELY successful and is more likely to capture SV's although novel genes will likely still be missed.

4.2 General steps

4.2.1 START: Pre-processed Sequencing Reads

De-multiplexed reads where individual bp calls have been made. This should include both the FWD and REV sequences of paired end reads if applicable. These reads may be long (pacbio) or short (illumina, sanger). Often this pre-processing will be performed by the sequencing platform/service provider. If this has not been performed then follow the service providers direction on cleaning and pre-processing your reads, this process is VERY specific to the sequencing platform.

4.2.2 QC:

Run all sequencing reads through some sequence quality metric tool. FastQC is common and easy to use. There are others which I shall not investigate significantly here. I used FastQC.

4.3 Quality Filter (if needed)

Hopefully there are not any reads that are significantly problematic. If there are specific reads that are bad this is a good time to toss them out. There are specific tools for this however many of the trimming tools discussed the next section can do this. Often the sequencing service will do this during de-multiplexing. All my samples had really high Phred scores so no sequences need to be removed purely based on phred score. They do need to be trimmed though....

Note: If there is significant GC content bias then you may need to drop or at least filter/compensate your sequences. Incorrect GC content (e.g. a bimodal distribution) indicates that you have over sequenced a specific, often repetitive region of DNA. It may be wise to filter out all the reads with abnormally high

GC content and see if they are all very similar in sequence and perhaps consider blasting them to see if your sample was contaminated or if they align with a specific region. Depending on the cause of the bimodality you can filter or remove some of these samples to create a theoretically more representative sample set. Ofcourse this is a nuanced art since there are many genomes and some may truly have a bimodal GC content distribution, this is very unlikely but you must employ your knowledge about YOUR organism to make an accurate determination. Ultimately you will likely be wrong no matter what you do... such is life.

4.3.1 Trim Low-quality ends:

Often the confidence in base pair calls is quite low towards the end of each read. Dropping the low quality BP is generally a good idea to make sure you don't have incorrect SNP calls. This can be done with *Trimmomatic*, *Cutadapt*, *BBduk*, *FASTP*, *Seqtk*, *NGSQC*, *SAMStat*, *FastQC* *htSeqTools*, *seqclean* others?

It may not always be clear if your sequences have been trimmed. Sometimes the service provider will trim your sequences (illumina tool does this quite well). Typically if ALL of your reads are exactly the same length (e.g. 151 bp for 150PE reads) then the sequences likely have NOT been trimmed. Trimming typically will not remove equal number of bp from every read and thus should create a left skewed normal distribution with an average that is 10ish bp shorter than the target read length (e.g. 140bp for 150PE). This 10ish bp decrease is mostly due to 5' bias trimming and not the adapter content bias trimming.

The trimming tool should perform 2 key tasks:

- Trip 3' Adapters. The sequencing tool will often "run-through" the end of the target dna and will sequence the adapter itself. This will cause all sorts of issues down the line. These 3' adapters should be trimmed, this is usually done by explicitly looking for the adapter sequence and then trimming EVERYTHING on the 3' side of that sequence. All sequence data after the adapter must be assumed as invalid and is tossed out. Hopefully only a moderate subset of the sequences need to have the 3' trimmed as this trimming can notably decrease your sequence k-mer length which isn't ideal. Less than may 20% needing to be trimmed is pretty great. You typically don't need to trim any ADAPTERS from the 5' end since the polymerase is running from the 5' end and starts slightly after the adapters. If you do have significant (any more than 1-5%) of reads with 5' adapter sequences then you should seriously question the sequencing technology and investigate. I have never seen this myself and frankly don't know what I would do.
- 5' Trimming. Just because you don't (or at least shouldn't) have 5' adapter sequences doesn't mean that you don't need to trim the 5' end. If you look at all the samples that are sequenced in a single "run" then the 5' GC content you WILL see a matching pattern of about 15bp for every sequence regardless of the true sequence identity. This is caused by many

reasons but often because of the fragmentation step. In theory the enzyme that chops up your dna does this at random.... however we know there are always biases so and may cut in a less than random fashion. There are many other reasons inherent to the sequencing methods that also drives some of this 5' bias. Regardless it is likely best to remove the first 10-15bp of all to remove this bias. Many trimming tools will handle this just fine.

There are multiple tools that can do your adapter trimming. Unlike alignment tools they honestly all perform relatively similarly, and which specific tool is often moot especially if you have access to a significant amount of computation power. Here is a short list of trimming tools and their benefits/drawbacks:

- Seqclean [**zhbannikovSeqyCleanPipelineHightthroughput2017b**]: An idea tool with lots of features and open source. This tool includes:
 - Removal of PCR-duplicates. During library amplification the same strand will be amplified multiple times which does not add any information. SeqyClean identifies two reads as "pcr-duplicates" if the first 35bps match. It's slightly more nuanced than that but thats the mostly how it do. PCR duplicates don't add information even to any CCNV analysis since in theory no two fragments should share an idnential first 35pb. That would mean that two different copies of the same chromosomes fragmented in the exact same way... Unlikely to happen more than once or twice and once or twice isn't enough to change the CCNV analysis!! Removing the PCR-duplicates helps reduce dataset size AND will reduce bias in later analysis steps.
 - In RNA seq data it will remove poly-A/T regions. But won't do anything in DNA sequences
 - Contaminant Detection and Trimming: If you provide a list of possible contaminants then Seqyclean will perform k-mer matching and remove any contamination reads (e.g. reads that match the sequence of the provide contaminants). Contamination detection is not required for my work thus this will be skipped
 - Atapter Id and Trimming (PE overlapping): Uses SSAHA [**ningSSAHAFastSearch2001**] algorythm adapter identification then will trim. Importantly in this step it combines paired end reads as well. In theory a paired end read should match perfectly in the middle with adapters sitting off on each reads 3' end. THis can create a high confident consensus read and sometimes will increase the length of the individual read. This step is essentially the same in AdapterRemoval and Trimmomatic!
 - Quality Trimming: Removes bp from the ends of a read taht are significantly lower than the rest of the read and that are lower than 20phred score. Has a rolling 50 and 10 bp window with a similar algorythum as Lucy.
- Trimmomatic: [**bolgerTrimmomaticFlexibleTrimmer2014**] An industry standard tool that is fast and effecient.

- Adapter Trimming: Requires a user provided list of adapter sequences to "remove" Can include any number of sequences here (pcr products, adapters, contaminants). Performs this trimming in one of two modes:
 1. Simple Mode: Super fast, each read is scanned from the 5' to 3' end any sequence matching a the user specified sequence. If there is an alignment everything on the 3' end is trimmed. The user can specify the number of accepted mismatches. This will find the targeted region anywhere in the read quickly HOWEVER it will NOT detect partial adapter sequences as is common in "read through" type errors.
 2. Palindrome mode: Will find the partial adapter sequence matches at the end of a sequence (common in read through adapter errors). It does this predominantly through aligning paired ends and removing the 3' overhang with some additional user provided partial sequence mapping checks. See their publication for details [bolgerTrimmomaticFlexibleTrimmer2014]. This is highly beneficial and can create much longer reads than otherwise possible! Works best IF you have paired end reads (as all modern sequencing should).
- Quality Filter: Two main quality filtering methods.
 1. Sliding Window Filtering: Standard approach. Scans from 3' to 5' and when the average quality of a sliding window of bases drops below a minimum threshold those bases are dropped. This way a single weak base won't cause an early truncation of the sequence while still removing the poor sequencing results towards the end of a sequence. This runs really quickly which is rad!
 2. Maximum Information Filtering: This method tries to balance keeping long reads while dropping bad bases. Really short reads are worthless however long reads with really poor phred scores are also mostly worthless. In theory there is some tipping point where a longer sequence with imperfect base calls is better than a shorter sequence with only high confidence calls. This mode tries to find this balance to "Maximize Information" within the read.
- Cutadapt: [martinAlgorithmsToolsAnalysis2014] A commonly used tool that is VERY effective with Illumina data. Requires detailed understanding of command line tools. Allows simple direct adapter searches and removals. Doesn't do much automatically but allows for specific tuning of each trim which can be beneficial. This has been updated and improved through at least AUG 2025! Cutadapt will perform:
 1. Adapter Alignment and Removal. I am confused by the way it aligns and identifies the adapters. However it does successfully do this on the 3' end!

2. Quality trimming. Drop low quality bp scores from either just the 3' end or from BOTH the 3' and 5' ends. Tries to balance keeping some bad scores that are surrounded by good scores.
 3. Will remove polyA in RNA sequences
- BBduk: [**BBdukGuide**] Developed by the Joint Genome Institute at LBNL in combination with DOE BBduk (Decontamination using Kmers) is a specific tool within the BBTools grouping. BBTools includes other mapping, masking, and merging tools that all work well together. This tool appears to be a state of the art cleaning tool with an impressive array of possibilities. It is hard to find information on the algorithm though.... Must consider. The more I read about BBduk (and the rest of BBTools) the more suspicious I become. They refuse to publish a paper or produce a solid comparison on anything but simulated data. Lots of forum posts about how great it is, and it is most certainly fast and easy to use.... but I care most about accuracy and I want it to be good but I haven't found a paper showing how good it is yet.
1. Quality-Trim Both ends
 2. Adapter Trim - Unsure if adapters must be specified or if a library of adapters are provided
 3. Force Trim Left and Force Trim Right: Can force trim the ends of your sequence for any reason. A common reason to trim the 5' end would be the GCAT is often not even at the beginning. Be careful though! Just cause the balance looks wrong doesn't mean that it should be trimmed. Mapping with a histogram should show if the first base pairs are frequently wrong or not. If they are not frequently wrong then don't remove those base pairs even if the GCAT frequency is off! Peep this thread post 12 <https://www.seqanswers.com/forum/bioinformatics/bioinformatics-aa/37399-introducing-bbduk-adapter-quality-trimming-and-filtering?t=42776>
- FASTP
 - Seqtk
 - SAMStat - might be a qc tool not trimming
 - htSeqTools
 - AlienTrimmer
 - AdapterRemoval
 - Skewer
 - Sickle
 - Ea-Utils

- Picard
- PEAR
- IeeeHom
- Lucy: Cant run paired FastQ
- Btrim: Cant run paired FastQ
- nesoni clip: A handle tool for trimming a specific number of bases of your sequence. This could be used to trim the first 10-15 bp of my sequence
- SO MANY MORE. At the end of the day there are so many tools and many perform very similarly. Best to just QC the results and make sure you have trimmed all the things that need to be and then just move on with life haha!

4.4 Genome Size Estimation, Read Quality Estimate:

NOTE: The method here is not the same as long read methods!! Short reads have slightly different statistics!

Helpful metrics should include: Histogram of read lengths, read depth, and others. Statistics for individual and cumulative samples should be listed. This can help us understand if a specific sequenced isolate had problematic sequencing which could explain any erroneous or wierd results!

FastQC I have generally used fast qc to check the quality of read both before and after trimming. Other tools work for this however I found fastqc was easy to use and had nice outputs!

Sequence Alignment:

Align sequence to reference genome. Sequence alignment is arguable the most impotant step of any whole genome analysis especially if phylogenies are to be created. Mistakes in alignment can cause quite significant changes in end results. The callange with sequence alignment is the inherent trade-off between mapping speed and the qulaity of that alignment. There is a fair argument that if you have access to a good computing cluster then sacrificing compute time for quality is recommended, do note, however, that just because a tool is slow it will not neccessarily have a higher quality. Additional consideration must be given to the characteristics of spp/genome you are trying to align. Low-complexity squences, repetitive regions, collapsed copies of sequences, contaminations, or gaps in the genome all present challenges to alignment and each tool deals with these issues in slightly different ways. Lastly the **REFERENCE GENOME** can have ENOURMOUS impacts on the end alignment quality. A low quality referenve genome with gaps, or improperly characterized repetitive regions will wreck havoc to any alignment tool with some tools fairing better to some types of issues than others [schilbertComparisonReadMapping2020]. Thankfull I have the high quality **near_complete** PacBio sequence.

Before we discuss the common tools used we must first mention the four primary indexing methods these tools use for multiple align.

1. **Burrow-Wheelers Transformation:** Developed in 1983 by David Wheeler and Michael Burrows the Burrow-Wheelers Transformation will reorganize a string of text grouping of similar characters together enabling faster character comparison. Originally used for preprocessing of strings before compression it has become very helpful for alignment algorithms. If you want to explore how to perform the BWT look at Advanced Data Structures: Burrows-Wheeler Transform. In practice the BWT is performed on the reference genome, simultaneously an indexing step is performed to create a map of each character in the BWT to the original position in the genome. This step can take some time but is only performed once. Once this BWT and index has been created we can take each individual short read NGS sequence and query the BWT for the first closest match (this is covered quite extensively by DB query scientists). The real magic is in how the algorithm handles inexact matching and gaps and is where most of the different BWT based algorithms differ (I think). I frankly cannot comment on how this process is done. Ultimately, however, the BWT based methods use very little memory after the indexing step has been completed and are extremely computationally efficient albeit with some potential loss in accuracy due to how inexact matching must be handled.
2. **Reference Genome Hash table Indexing:** NOTE: I am not expert on this and I have not researched the details of this process significantly. This is my current understanding but you should use other resources if you want to understand the details of these processes.

Similar to BWT hash table indexing is a database indexing method designed for faster database queries. I am unsure when this was first developed but hash tables have been around for a VERY long time. Functionally instead of representing a genome as a whole string the genome can be "indexed" into a hash table where each value in the hash table is a list of length k (also known as a k -mer). These k -mers are of some preselected length k (the specific length is actually super important). Typically there is some "offset" array of k -mers that represent different starting positions of a potential k -mer which points to a specific position within some position array where the full sequence is listed. The k -mers within the offset array can overlap significantly and contain the same information multiple times in different ways which helps with the oligomer comparison later. Once this hash map has been created the piece of "target" DNA that is to be aligned is then broken down into overlapping "target" k -mers and queried against the previously created hash index. When matches are found the "seed location" aka the locations where the short target k -mers are recorded. Up until this point, as I understand, many hash index based systems operate very similarly. There is some deviation with long read sequences since theoretically the long read is known to go together but the basic fundamentals are approximately the same.

Once the seeds have been determined a plethora of possible algorithms are used to examine all the possible seeds for all provided "target" k -

mers and find the overall "best" alignment. Best is arguably subjective and much care should be used when deciding which algorithm you use for a specific species, some are better at expecting large structural variants others are better at handling small SNPS, some are better for short read (and thus short k-mer) sequences and others are better for long read (long k-mer) alignment.

A note on k-mer length. k-mer length is CRITICAL. Theoretically we need each k-mer to be wholly unique within the genome. If k is too small then there will undoubtedly be repeated k-mers within the hash. The Highly repetitive regions in many genomes cause a significant issue here and can easily require $k > 1000$ to have confidence in sequence alignments unless repetitive regions are dropped. Large k-mers however will be less sensitive which is not ideal. There are also two hard limitations to choice in k-mer length. 1. k cannot be longer than the sequence fragment size. E.g. if you sequence on illumina PE150 then $k \leq 150$ we cannot create data from nothing. This specific limitation is a driving factor for longer read sequencing methods like PacBio and Nanopore. 2. Large k values present a SIGNIFICANT compute challenge, the RAM required to store the hash index increases linearly with k and genome size. For a human genome of 3 billion base pairs and a $k = 100bp$ the hash table would easily need 75GB of RAM ($RAM = GenomeSize * k / 4$) and that's assuming minimal overhead RAM costs for the hash. Of course there are ways to obtain this amount of RAM and the hash algorithm can be tuned to reduce RAM overhead but it is still a notable problem. Choice of k-mer length is not trivial and can change results and compute time dramatically. Choose k-mer carefully.

3. **Hashing Reads:** Theoretically similar to hashing the genome. Unsure of the nuances here
4. **Sorting and Mergin:** Unsure the implications of this method. Appears to be significantly less common than the first two. Includes Slider and Syzygy apparently [muFastAccurateRead2012]

TLDR: Fundamental methods of alignment: BWT is a fast efficient relatively accurate method and is especially suited to short read sequencing. Hash table indexing is memory intensive and slow but usually provides the highest accuracy and can detect large genomic rearrangements and other similar genomic phenomena more easily than many BWT methods. HOWEVER, because each of these methods depend on many pre and post processing steps outcomes of any one tool VERY dependent on the specific tool. Do not select a tool simply because it is a BWT or a Hash table index, select a tool because it performs with your system and sequencing platform. [liDiminishingReturnIncreased2014][wuBitpackingTechniquesIndexing2016][misraAnatomyHash]

Lastly: Just because a tool works REALLY REALLY well for the human genome doesn't mean it will work for your sample type. From a genomic phenomena POV the human genome is pretty darn simple

and many tools designed for human genome sequencing will fail to capture the true nuance of your system

Note: There are multiple metrics for how well some sequence aligned to the reference genome.

1. **Mapping Percentage:** The percentage of reads that were mapped to the genome. Higher is generally better. Above 90% is probably good. Importantly just because a read was mapped to the genome doesn't mean it was mapped to the right spot. This presents a significant challenge.
2. **Other Metrics:** There are countless other metrics one may be interested in, however, genetic sequencing presents an unworkable problem. Rarely do we know what the "right" answer is so although we can assess each tool's sensitivity and accuracy with respect to some ground truth we have no truth to ground. Many groups have created simulated genomes to test these tools and have provided quite interesting and helpful comparisons for how these tools are likely performing. Importantly it has been noted that the performance of each tool changes based on the specific organism (or at least kingdom) that is being sequenced. In my opinion there is enough nuanced variation in these studies that if you need the utmost confidence in your results then you must either find such a study that has been performed on your system of interest or perform such a study yourself! This is likely important if you know your organism presents some rather unique methods of inheritance or mutation. If this is not the case then just choose any of them and they will probably give you a decent answer. Better yet use a few tools and if they all agree then that answer is probably as close to right as you will get. Do also think about your final application and what specifically you are trying to learn, that can guide you if you want a more accurate or more sensitive tool.
3. **I am partial:** Currently I am moderately inclined to use Novoalign, BWA-MEM, and STAMPY paired with the GATK variant caller. They seem to have a good balance of sensitivity and accuracy in most tests I have read.
4. Unfortunately, It appears in most cases sensitivity and accuracy are inversely related.

Here are a few of the existing alignment tools and some commentary on their best uses:

1. Burrow-Wheeler Transformation

- **CLC-mapper:** CLC-mapper is developed by Qiagen. As such it is a proprietary software suite that is primarily used through a graphical user interface. This significantly limits its use on compute clusters and as such, in my opinion, is not suitable for large scale genomic analysis (admittedly I have never used this tool). Regardless according to [schilbertComparisonReadMapping2020] it does not

provide the same quality results as novoalign or BWA-MEM when used to align plant genomes (human genomes are more simple and perhaps CLC-mapper is better for these more rudimentary genomes) (yes I know I am throwing shade lmao - go work with a non-"model" genome and then come talk to me. Human genetics are easy).

- **SOAP2:** Uses Burrows-Wheeler Transformation indexing algorithm. [schilbertComparisonReadMapping2020] [wuBenchmarkingVariantIdentification2019]
- **Bowtie2:** Based in the burrow-wheelers transformation Bowtie2 is an excellent longer read illumina style sequences around 150-500bp (Bowtie2 is not particularly suited for modern long read sequencing e.g. pacbio or nanopore). Bowtie2 uses a remarkably little memory and is not particularly core intensive making it an ideal platform if you do not have access to a compute cluster. It is generally more sensitive than many BWT counterparts such as BWA (Burrow-wheelers alignment) and SOAP2. I have found that more than 36 cores will not particularly decrease run times. Be sure to set proper setting for Paired End sequences and non-paired end sequences! You can find run code and some associated README in this git project! [langmeadFastGappedreadAlignment2012a] [schilbertComparisonReadMapping2020]
- **BWA-MEM:** An update on the older BWA tool. This tool regularly comes up as one of the best BWT based alignment softwares. Combined with GATK BWA-MEM should be fast and give really good results for short read sequencing methods.
- **BWA-SW:** Finds local alignments of the reads rather than global alignments.
- **GEM3:** [schilbertComparisonReadMapping2020]

2. Reference Genome Hash Table Indexing:

- **novoAlign:** Optimized for NGS data. Unsure if it works for long read sequencing I doubt it. Harrolded as one of the most accurate alignment tools NovoAlign is based on hash table indexing. Full details of how this system works is a bit hard to find. It appears to be proprietary to some notable degree. Should be free to institutions [schilbertComparisonReadMapping2020]
- **GSNAP:** Uses hash table indexing. [wuBenchmarkingVariantIdentification2019]
- **SNAP:** Uses large continuous seeds
- **BFAST:** Uses large spaced seeds to improve sensitivity
- **GASSST:** Uses a large array of filters to improve alignment speeds
- **SHRiMP2:** Uses large spaced seeds to improve sensitivity
- **SeqAlto:** Uses large continuous seeds. Is only suitable to sequencing methods where reads are greater than 100bp. Published here Fast and Accurate read alignment for resequencing [muFastAccurateRead2012]

3. **Read Hash Table Indexing:** I don't quite understand how this differs/is better/worse from the Genome Hash Table indexing. Not super common though. Supposedly helpful when the reference genome is small (e.g. a transcriptome) [**muFastAccurateRead2012**]

- **MAQ:**
- **SeqMap:**

4. **Sorting and Merging:**

- **Slider:** Requires a large fast RAM
- **Syzygy:** Requires a large fast RAM

5. **Other/Hybrid:**

- **Stampy:** A fast yet sensitive alignment algorithm that uses a hybrid mapping algorithm (hybrid between a hash and the burrows-wheeler method) to be more sensitive than the burrow-wheeler transformation counterparts (bowtie, bowtie2 ELAND, BWA) but faster than the high sensitivity hash based systems (Noalign, Mosaic). Stampy is particularly well suited to short read Illumina sequencing. Link to Stampy Git <https://github.com/uwb-linux/stampy> [**lunterStampyStatisticalAlgorithm2**]

6. **Unknwon:**

- **Geneious Prime:** Proprietary tool (\$200 yearly). The Geneious alignment tool is easy to use and in my experience fairly accurate. FIND PAPER COMPARING ACCURACE TO OTHER TOOLS. The although GUI interface is convinient it severely limits the use of compute clusters (unless there is some geneious command line tool that I am un-aware of...) FIND THE ALGORITHM TYPE IT USES. In my experience it is substantially slower than Bowtie2 but appears to provide more accurate results and handles SVs slightly better. GEneious prime software is quite nice when trying to work with genetics datasets and undestand what you are looking at so I do recommend the software in general. However if you are trying to align more than just a few sequences then I would likely avoid this tool.
- **BBmap:** Need to investigate but advertized as ideal for highly mutated and changing genomes.

7. **Old or deprecated tools:**

- **Bowtie:** The older brother of Bowtie2, bowtie is really only good for very short reads (≤50bp). Though a historically fantastic algorithm newer algorithms have both a higher sensitivity and increased speed.

- **BWA:** The old version of BWA. Careful of the exact name of BWA and BWA-MEM, many software tools were benchmarked against "BWA" and it is not always specific if they benchmarked against the most modern version "BWA-MEM". Frankly, neither BWA nor BWA-MEM is the best tool for my application so the issue is largely moot!

Some advertised "alignment" tools should not be used in this short read sequence alignment steps:

1. **Mauve:** A powerful tool for multiple alignments. Mauve is designed to detect whole genome rearrangements and is best used for comparative genomics. Aka taking two assembled (likely de-novo assembled) genomes and comparing them! Helpful for answering questions about synteny, or identification of large SVs. This can be an excellent tool for multiple alignment prior to building a phylogeny. It cannot handle short read sequences. Also note that Mauve has a function (if rudimentary) GUI which makes it slightly less convenient to run on a cluster. I believe Mauve alignment is possible using only command line tools and associated run scripts however I have not yet attempted this (Jan 2026).
2. **LastZ:** A good tool for long sequences LastZ is more similar to Mauve than the tools we need for this step. LastZ is an excellent choice for multiple alignment prior to building a phylogeny!

Discussion on the tools I plan to try: Currently it is not possible to know which tool will give me the best results thus I hope to try: **BWA-MEMs, STAMPY, bowtie2, Novoalign, and maybe SeqAlto.** With the known CCNV in Bd I have concerns of repetitive areas and large structural variants so I want something that can handle these variants well, I believe that Novoalign, STAMPY, and SeqAlto are particularly suited to these challenges. Most of the NGS data I have from Imani are PE150 or larger so having a longer k-mers on the order of 28 or so will give me decent coverage without sacrificing specificity. For some of the downloaded historical reads I should decrease the k-mer length notably. In my case I don't really need to worry about memory since I have access to the cluster and the Bd genome is pretty small. I am particularly inclined to use STAMPY because it has been used with good results in Bd in the past and according to Mu et al. 2012 [**muFastAccurateRead2012**] Stampy dealt with indels, SNPs, and MNPs very remarkably. Novoalign has also been updated since then and has some really good reviews despite the claim that it can run quite slow. Lastly I want to use BWA-MEMs because it has been used a good bit in Bd genetics research and has proven itself accurate. Bowtie2 is on the list because it is easy to implement and fast to run. It would be nice to know if bowtie2 was adequate since using that tool would save other the required use of a cluster.

NOTE: Although I tried to make sure I cited appropriate journals/publications in this list. The following publications were all extremely helpful for identifying

and understanding the different tools. [kimBeginnersGuideAssembling2022], [rosenblumComplexHistoryAmphibiankilling2013], [muletz-wolzDiverseGenotypesAmphibiankill], [nielsenDiploidyHaploidGenus2021], [byrneCrypticDiversityWidespread2019], [refsniderGenomicCorrelatesVirulence2015], [lunterStampyStatisticalAlgorithm2011], [langmeadFastGappedreadAlignment2012a], [kimBeginnersGuideAssembling2022], [wuBenchmarkingVariantIdentification2019], [liDiminishingReturnIncreased2014], [wuBitpacking], [muFastAccurateRead2012], ([manconiRemovingDuplicateReads2016])

4.5 Polishing Genome Assembly (PacBio or Nanopor primarily):

Okay now that we have assembled the genome properly. WE might want to polish the genome a bit to improve the results a little bit THis polishing is primarily needed in long read sequencing applications where the sequencing method has higher BP error rates. Upon reading more I do not think that traditional Illumina NGS piplines really need this genome polishing step. These NGS piplines are more susceptible to large sclae errors not small scale errors which this polishing really shines at. Regardless there are many tools for this. Here I will list a few of the short read sequencing based tools:

1. **GCpp**: <https://github.com/PacificBiosciences/gcpp>. This appears to be primarily for PacBio sequences....
2. **Pilon**: <https://doi.org/10.1371/journal.pone.0112963> THis will both correct/clean draft assemblies and do sequence variant calling.
3. **Racon**: Some sources claim this is form Short REad sequencing however it appears to be more geared for PacBio Sequences if you ask me <https://doi.org/10.1101/gr.214270.116>
4. **Inspector**: <https://link.springer.com/article/10.1186/s13059-021-02527-4>
5. **CONSENT**: <https://www.nature.com/articles/s41598-020-80757-5>

SAM file reformatting: Sam files coming from alignment tools are often not structured in a way that is super helpful. I have found that they often need:

- Addition of reading groups, this is sometimes needed when multiple replicates of the same sample are sequenced on different Illumina runs. Not applicable for me but it is important in some systems.
- Coordination sorting of reads. When aligned each read isn't really moved around in the file. Each read is simply given a location that it belongs in. It can be helpful to reorder all the dna reads so that they reflect some standard pattern. I think if you include the reference genome it orders it like the reference genome is ordered!

I did all of this using Picard which might use SAM tools under the hood. Really not sure. I am only going to use Picard though since it is working and neither of these clean up steps are particularly variable.

4.6 Duplicate Marking and Indexing:

We have multiple reads now that probably look pretty decent. However when our library was created we it was likely PCR amplified. This amplification theoretically should amplify a region MORE than other regions which may cause artificially increased read depth of those regions. This may be an issue for CCNV analysis and for phylogentic analysis as these repetitive regions may throw our results. FOR CCNV analysis it is averaged out across a chromosome so a single over represented region really isn't an issue. It is more of an issue for CNV situations though. Regardless we probably should remove and mark duplications before we proceed with additional analysis.

There are a few tools that are used for this. They generally land in two groups [manconiRemovingDuplicateReads2016], [pingHowErrorCorrection2025]. Alignment-Based deduplication tools and alignment-free deduplication tools. Some work for paired end and some work with single end. In general the alignment based tools are a bit faster however they can be biased by the specific short read mapping tools used to align the sequences. These alignment-based tools match both the position and the sequence identity to identify duplicates. The alignment-free tools, however, only compare each alignment to all the other alignments independently! This is great however it is computationally costly and can take forever! Lastly these tools also have a bit of variability in the identification of "duplicates" e.g. sometimes they look for nearly-identical or purely identical sequences. This is important to think about when deduplicating as your organism might change what you are interested in using.

NOTE: Deduplication is not the same as repeat identification or transposable element identification. Deduplication is the attempt to remove method induced duplications! It is tricky though, because a true repeat can look similar to a duplication.

1. **Sam Tools rmdup:** Alignment based, single and paired end, Nearly identical. does NOT allow for read correction and nor allows mismatches for paired end
2. **MarkDuplicates Picard:** Used by Rosenblum et al. 2013. Alignment based, single and paired end, Nearly identical, Does NOT allow for read correction and nor allows mismatches for paired end
3. **SEAL:** Alignment based, paired end, Nearly identical
4. **FastX-Toolkit Collapser:** alignment-FREE, single-end, identical
5. **FastUniq:** Alignment-free, paired-end, identical
6. **Fulcrum:** Alignment-free, single and paired end, nearly-identical
7. **CD-HIT-DUP:**Alignment-free, single and paired end, nearly-identical. Does read correction and allows mismatches for paired end
8. **GPU-DupRemoval:** Alignment-free, Single and paired end, nearly identical

9. **SEED:**
10. **pRESTO:**
11. **ParDRe:**
12. **MarDRe:**
13. **PCRdupliactes:**
14. **NGSReadsTreatment:**
15. **Nubeam-dedup:**
16. **BioSeqZip:**
17. **AmpliCi:**
18. **Minirmd:**
19. **Fastp:** Alignment-free, Does read correction and allows mismatches for paired end

There is another newer class of deduplication methods as well. These depend on the UMI approach (unique molecular identification) where each PCR amplification reaction adds a unique identifier to each amplified region. This provides unique IDs to all sub sequence components and helps reduce the duplication errors in shotgun sequencing. This must, however, be performed during the library prep. Here are some of the tools that have been created for UMI based deduplications:

1. Je - Is based on the Picard system
2. UMI-Reducer
3. AmpUMI
4. zUMIs
5. Umitools
6. Calib
7. UMICollapse
8. UMIC
9. DAUMI
10. Gencore
11. markdup(Samtools)

Upon reflection on all of this. And recognition that my system is likely highly repetitive with frequent large structural repeats in the form of CCNV I think I want to use a deduplication system that allows no mismatching. That being said I do want to run a few tools so I will point towards these:

1. **MarkDuplicates PICard:** Was used by Rosenblum et.al.
2. **CD-HIT-DUP:** Does read correction and allows for some mismatching
- Will reduce to only using PICARD and SAMTOOLS for simplicity sake
3. **SamTools rmdup:** Commonly used tool that appears to perform well.
4. **MarkDuplicatesWithMateCigar** An update on markduplicates (picard) but with the mate CIGAR attachment. It changes the way it breaks ties. I am currently unsure exactly how it is different from the older markduplicates tool but it is supposedly far superior.

Last note: Some people argue that a PCR error-correction step should be performed prior to deduplication. This theoretically would account for mistakes in the PCR amplification steps of library prep and then more accurately mark duplicates. In my opinion this at best does little and may actually introduce additional mistakes reducing the observation of genuine unique DNA sequences. As such this step will not be performed.

4.7 Alignment statistics

: Now we have an aligned sequences and we have organized things well and have marked all the duplicates. However we still have it as a nice looking SAM file. We should create some statistics on these files so we can get a sense for how our alignment worked. Perhaps the best tool for this is SAMtools. Which makes all sorts of lovely alignment centric metrics. It won't give us the typical N50, N60 etc metrics we usually love but we need to create the consensus sequence prior to that. And we will lose some info if we do that! So lets use sam tools. Here are the list of samtools I think I want to use:

1. **samtools stats:** Creates a detailed report of the whole genomes. Summary numbers which will be helpful, quality scores, gc content, coverage depth, and good readdepth distributions.
2. **samtools idxstats** This will give us chromosome or contig level metrics. Including read depth etc. This read depth will give us insight into the CCNV of each chromosome. I still will need to run the relative snp frequency analysis (something that GenomeScope2.0o does on a whole genome but perhaps not on individual chromosome level) but this will give us a sense.
3. **samtools coverage** This will give us individual contig coverage depths. Probably not needed if I do idxstats

4.8 Consensus Sequence:

We have a some SAM and BAM files that are looking great. However It would be good to have a consensus sequence of it all tbh right? Usign sam tools it should take the aligned fragments and create a consensus sequence that tells us how everythign looks! This will format it as a FASTA or FASTQ file depending on settings. There are other tools that can do this but from what I understand they won't make much of a difference in the end. Do note, however, that for the CCNV analysis I will need to use the base SAM alignment files otherwise I will not get the right snp frequencies.

4.9 Final Assembly Assessment: (question: Is this step necessary for a genome resequencing and build from a reference genome? Theoretically this should be enough so long as our sequences had relatively good alignment with the reference genome. Ponder.)

Now that we have assembled a genome, polished it some, and then deduplicated our reads. We should take a moment and assess our genome for quality! There are two primary areas to peak at. 1. We should look at some generic statistics of the genomes such as N50, NG50, L50, LG50, NG50, LA50, LA50, and NA50 etc. 2. We should also look at our assemblies for completeness and contamination. Something of note here! Many of these tools look for equal balance of contig depth etc as you see with traditional diploid genomes and thus may throw errors for us examining CCNV. As such I recommend "GenomeScope2.0 which is specifically designed for polyploid genome assemblies).

Here are some generic tools for assembly assessment looking at fundamental statistics:

1. **assembly-stats**: <https://github.com/rjchallis/assembly-stats>
2. **QUAST**: <https://academic.oup.com/bioinformatics/article/29/8/1072/228832?guestAccessKey=>
3. **GenomeScope**: <https://academic.oup.com/bioinformatics/article/33/14/2202/3089939?login=true>
(Not to be confused with the 2.0 version of the same program which is ideal of rpolyploids)
4. **kat**: <https://academic.oup.com/bioinformatics/article/33/4/574/2664339?guestAccessKey=>
5. **Squat**: <https://link.springer.com/article/10.1186/s12864-019-5445-3>
6. **GenomeQC**: <https://link.springer.com/article/10.1186/s12864-020-6568-2>
7. **BlobToolkit**: <https://academic.oup.com/g3journal/article/10/4/1361/6026202?login=true>

Here are some tools that look at completeness and contamination:

1. **CheckM**: Specific for microbial genome assemblies
2. **BUSCO**:
3. **PhyloOligo**;
4. **LTR_{retriever}** : **Mercury**;
5. **GenomeScope2.0**: Is specific to polyploid. GenomesScope2. and SMUDGE actually look like quite handy tools. It uses raw reads to build ploidy estimates (among other stats) about the whole genome. It is specifically designed for polyploid systems however it is not compatible with aneuploids. I wonder, however, if I can give it filtered contig specific reads for it to then run on.... Something to think about.

4.10 Choices

1. **Option 1: Genome Annotation and partial genome phylogenies/variant analysis** If you want to annotate the genome and compare small regions of the genomes of multiple individuals to each other. Or do other SNP and or variant based analysis go to this section. Here broadly we will do SNP identification and INDEL calling. As well as annotation of the whole genome!
2. **Option 2: Whole Genome comparative analysis and phylogenetic trees or PCAs.** If we want to do a WHOLE genome (or even partial mitochondrial genome type analysis) where we will compare whole regions and see the relatedness of these individuals we don't really need to do SNP and indel calling. Importantly we do need to have performed some sort of duplicate masking or removal of duplicate sequences. This will help make the phylogenetic analysis tool run more quickly which is nice and help be more explicit on how these repeats are being handled. The different tree tools don't always do this the same way.

4.10.1 Option 1: Genome Annotation and partial genome phylogenies/variant analysis:

4.10.2 Variant (SNP and INDEL) Calling:

THIS section is rather broad since multiple things need to happen here. Fundamentally we need to identify the locations where our new sequences are DIFFERENT from the reference sequence as well as remove duplicate reads (or mark them at the very least). THIS can be done with many tools but it is essential for identifying where everything is varied! Coming out of the alignment tools we simply know where all of our paired end reads land and how they are organized. WE don't have any idea on how they similar or dissimilar they are to the reference genomes. Or at least not any beyond the the percent aligned metric that is given.

There are a few different things we need to look for Here

1. **SNP:** Single Nucleotide Polymorphisms. Aka locations where single nucleotides are different from the reference genome.
2. **Indels:** Insertion or deletions of DNA. Sometimes this is only a few base-pairs and sometimes this is MANY. But we need to identify these!
3. **SV:** Structural variants. Essentially just indels that are greater than about 50bp in length. These are generally more rare and actually might not be well identified in our Illumina data. OUR short read sequencing only has reads of about 150bp so a 50bp change would likely cause that read to not pair super well.

There are multiple possible tools for these tasks. A short list includes:

GATK: Used by rosenblum et al. Industry standard

DeepVariant: Google based AI tool. Don't know much about it.

Strelka2: Probable best for smaller datasets. Fast

FreeBayes: Bayesian based model

VarScan2:

NextGENe:

4.10.3 Option 2: Whole Genome comparative analysis and phylogenetic trees or PCAs.

If we want to do a WHOLE genome (or even partial mitochondrial genome type analysis) where we will compare whole regions and see the relatedness of these individuals we don't really need to do SNP and indel calling. Importantly we do need to have performed some sort of duplicate masking or removal of duplicate sequences. This will help make the phylogenetic analysis tool run more quickly which is nice and help be more explicit on how these repeats are being handled. The different tree tools don't always do this the same way.

5 PacBio Analysis

5.1 Generally pacbio analysis notes

: Creating a de-novo assembly from long read PacBio sequencing is a really great idea! We have the data and while performing the de-novo assembly we are more likely to identify SV's and new genes. Since we already have a hifi pacbio sequence I recommend doing a hybrid type assembly where we do a de-novo assembly and then scaffold this genome to a good reference genome. At which point we can make our structural variant calls and gene annotations (see a Beginners Guid [**kimBeginnersGuideAssembling2022**]).

5.2 Haplotype level assembly tools for pacbio sequencing:

See below in notes section for hifiasm methods of assembling pacbio sequenes. Hifiasm is easy to use however it is really only optimized for diploid organisms and isn't a great candidate for our application. I was able to run it successfully and it identified a primary and secondary chromosome for each cromatig assembly howevber I want a higher level.

Typically if you are doing haplotype level assemblgly you need HiC which will tell us how bits of DNA connect to each other. Howevber I do nhot have HiC and do not think i coule even get it if I had the funds. I think, however, that I might can get the assembly I need without HiC becausxe of the rather absured read depth that i have gotten.

Haplotype level assembly without hic may be really tricky. However I think there is a way to do this if I am clever. I have a good reference genome and I have really really deep coverage of each cell. Instead of doing a denovo assembly:

1. Map hifi reads to the referenve genome. a. this should allow me to get all sorts of metrics and separate out each read to their own little corner. I can then treat each subset of reads that are associated with a specific chromosome as it's own problem to be solved.
2. Estimate: a. Copy number b. allele-frequency (similar to what is done in ngs for ccnv) this shoud give me the stuff I need for my CCNV projecr c. Structural variantgsx d. heterozygosity e. at this point I could also probably drop the mitochondrial genome information. But maybe not. gтетting mt count couyld be neat idk f.

Letsdiscuss some denovo assembly tools for Pabio Sequencing.

1. hifiasm
2. HiCanu
3. Verkko
4. LJA
- 5.

6 Population and intra population Phylogenetics

I will discuss at length the idea phylogenetics in this section. I challenge the reader and myself, however, to question if phylogenetics is even the end goal of my work here. My goal with these samples is to better understand the similarity of the different strains and gain some insight into the microevolutionary history of these samples. I want to learn how related and from which pond did these strains perhaps spread. This sounds like phlogenetics but it is perhaps not. Ultimately all I may want is a single metric that classifies the "similarity" of these strains, which of its self is likely an impossible task.

Doing an population phylogenetic and intra population phylogenetic analysis is not as trivial as one may think. Infact I may argue that pupulation and intra population phylogenetics may need an alternative name and might not be phylogenetics at all. To quote an email to me from Prof. Jesus Martinez-Gomez:

"Typical population level data violate assumptions of standard phylogenetic methods (which are underlied by nucleotide substitution models). Phylogenetics assumes strict bifurcation, no introgression (depending on how you analyze data) and that 'species' have no recombination (variants are fixed at species level). Population genomic methods can be used to generally capture microevolutionary processes inferring ancestry, in the same way as phylogenetics, is a bit tricky. But since you are a clonal organism then I think assumption of phylogenetics might work."

He did note, however, that in a clonal organism the fundamental assumptions of phylogenetics might still apply.

Needless to say I find myself in dire need of a deeper understanding of the dark arts of phylogenetics. As usual, what I had assumed to be a relatively simple comparison of genetics closness inferred by the relative sinmilarity of the genomes is not simple, relative, or close to my comprehension at all. Thus let us go on a journey together to learn the underlying fundamentals of phylogenetics:

6.1 Phylogenetic Fundamentals:

6.2 Population phylogenetic fundamentals:

6.3 Phylogenetic tools:

7 Imani Sequences What I have done (in rough order with dates)

7.1 Genius Stuff

I did some random things in Geneious including making some alignments to the genome and some rough phylo trees. I hadn't cleaned anything yet though so that was bad!

7.2 Bowtie alignment 1

Aligned stuff with bowtie using the cluster. HJadn't cleaned stuff yet though so that was bad

7.3 Illumina Sequencing Analysis Real:

- (2026-01-21: Fast QC) Analyzed the sequences (imanis and downloaded) with FastQC to asses the common issues. Most are pretty okay.
 - Imani's sequences
 1. Per Base sequence content: Most Failed. The first 15ish BP of most sequences have a poor pattern of relative GC content. This pattern almost perfectly aligns with each sequencing run (5720 and 5912). All the 5720 look the same and all the 5912 look the EXACTLY the same except for 5912-JEL272 which looks somewhat like the other 5912 samples but not exact, this may be driven by its POOR GC graph which is notably bimodal.... Discussed later ULtimately I believe that the first 15ish BP should be trimmed b/c there is so much consensus between all the fragments that either it is a fragmentation bias or a bias of the specific sequencer. I suspect it is a sequences (perhaps adapter) based bias since the per bas sequence content is paired between each run.

Be careful though! Just cause the balance looks wrong doesn't mean that it should be trimmed. Mapping with a histogram should show if the first base pairs are frequently wrong or not. If they are not frequently wrong then don't remove those base pairs even if the GCAT frequency is off! Peep this thread post 12 <https://www.seqanswers.com/forum/bioinformatics/bioinformatics-aa/37399-introducing-bbduk-adapter-quality-trimming-and-filtering?t=42776>
 2. Adapter Content: The adapter content for all samples is rather high for the nextera transposase sequence. Perhaps 20% at position bp position 140. This is likely because of there hasb't been any adapter trimming. That must be trimmed! Which is my next step. This trimming thing begins sometimes as soon as 50-60bp. This isn't ideal but not that big of a deal if this is relatively rare

in the sample. This is caused by sequence "read-through" and really isn't so bad so long as the 3' end is trimmed. Ultimately it is caused by inconsistent fragmentation which is kinda common in nextera kits. PolyG and polyA are other common artifacts in illumina sequencing and can/should be trimmed. If it's only a small fraction of the reads it shouldn't cause any issue at all.

3. Deduplication Level. All samples have a non "ideal" deduplication level. I am not concerned because of bds weird CCNV genome I expect there to not be even coverage across the genome.
4. 5912-JEL272: Bimodal per sequence GC content. Should be normally distributed around like 40% or so.
5. 5912-JEL216: Bimodal per sequence GC content. Should be normally distributed around like 40% or so.
6. I may need to toss JEL272 and JEL216. The bimodal GC distribution has me sorta concerned. But so does the distro of the published genomes so IDK. Might isolate the high GC% sequences and blast those to see if they align with anything.... Hard to say.
7. Average Length: All samples are VERY consistently 151bp in length. This is great means the machine didn't have any systemic issue. This consistent read length will change once I trim but that is to be expected. Lets trim.

– Published sequences

1. Hello world

- (2026-01-27 Trimmomatic:) I ran Trimmomatic version 0.40 (most recent of Jan 2026) on all of Imanis samples. I used my bd genetics analyses scripts specifically the run files within the Trimmomatic folder of this git branch.

```
# Github link: https://github.com/malloc3/bd_genetics_analysis.git
# commit 2bd99208b569fa5eb02994aa11b892a83cf0aa78 (HEAD -> main, origin/main)
# Author: Cannon Mallory <mallorycannon15@gmail.com>
# Date: Tue Jan 27 19:33:04 2026 -0800
#
# editing the error and outlog names
```

The Specific settings used for the trimmomatic command were

```
java -jar "./Trimmomatic/target/trimmomatic-0.40.jar" PE -threads $numb
-trimlog "$trim_log_file" -summary "$summary_file" -compressLevel 9 -ve
"$full_input_file_path_1" "$full_input_file_path_2"\
"$full_output_file_path_1_paired" "$full_output_file_path_1_unpaired" \
"$full_output_file_path_2_paired" "$full_output_file_path_2_unpaired" \
```

```

HEADCROP:15\
ILLUMINACLIP:Trimmomatic/adapters/NexteraPE-PE.fa:2:30:10 \
LEADING:3 \
TRAILING:3 \
MAXINFO:150:5 \
MINLEN:36

```

```

# HEADCROP:15 removes the first 15 bases without question (I think I need)
#ILLUMINACLIP: NexteraPE-PE.fa: :<seed mismatches>:<palindrome clip threshold>
# <simple clip threshold>:<minAdapterLengthPalindrome>:<keepBothReads>
# ## NexteraPE-PE.fa - is the folder that has the adapters we want to use
# The specific format of this folder is IMPORTANT beyond the file name
# ## seedMismatches: specifies the maximum mismatch count which will be used
# ## palindromeClipThreshold: specifies how accurate the match between the read and
# ## simpleClipThreshold: specifies how accurate the match between the read and
# ## minAdapterLengthPalindrome: (optional int) specifies the minimum adapter
# ## keepBothReads: (optional boolean) specifies if both reads should be kept
#
# MAXINFO uses their special algorithm to maintain maximum read length
# Its clever and should do good for us
# Remove leading low quality or N bases (below quality 3) (LEADING:3)
# Remove trailing low quality or N bases (below quality 3) (TRAILING:3)
# Scan the read with a 4-base wide sliding window, cutting when the average
#
# Drop reads below 36 bases long (MINLEN:36)

```

- **SeqyClean:** Ran seqy clean on all samples. Need to perform post clean qc. I think I probably messed up the 5927 samples because (at least according to trimmomatic results) it used both the Nextera adapters and some Illumina universal adapters. Will likely need to do another trim on that group.
- **FastQC of Imains trimmomatic trimmed samples 01/28/26**
 - All samples retained over 95% of all reads with most retaining 98% to 99% of all reads post trim.
 - The first 15bp were trimmed which made the per bp gc content ratios look way better. Still undecided if this was completely necessary. But it does make the qc better. We shall see if it assembles better though.
 - All samples have really clean histograms. All samples have skewed lengths due to trimming so that's normal.
 - Many samples have a flag on deduplication. I am not worried because the Bd genome isn't even b/c CCNV

- All samples from 5720 pass all QC check with flying colors.
- Samples from 5927 have too much Illumina Universal Adapter Sequence. I should re run the trimmomatic with these adapters included for trimming. I think I will just run it again from the original files I think.
- **Trimmomatic redo of 5927 samples 01/29/26:** Same as before however this time I will use the TruSeq3-PE.fa adapters and not the nextera. I should have done that in the beginning. Based on the first Fastqc you can see in the adapter content that the 5927 had different adapters than the 5720 samples. New settings are.

```
java -jar "./Trimmomatic/target/trimmomatic-0.40.jar" PE -threads $numb
-trimlog "$trim_log_file" -summary "$summary_file" -compressLevel 9 -ve
"$full_input_file_path_1" "$full_input_file_path_2"\
"$full_output_file_path_1_unpaired" "$full_output_file_path_1_unpaired" \
"$full_output_file_path_2_unpaired" "$full_output_file_path_2_unpaired" \
HEADCROP:15\
ILLUMINACLIP:Trimmomatic/adapters/TruSeq3-PE.fa\
LEADING:3 \
TRAILING:3 \
MAXINFO:150:5 \
MINLEN:36

# HEADCROP:15 removes the first 15 bases without question (I think I ne
#ILLUMINACLIP: TruSeq3-PE.fa: :<seed mismatches>:<palindrome clip thre
#
#       <simple clip threshold>:<minAdapterLengthPalindrome>:<keepBothR
#   ## NexteraPE-PE.fa - is the folder that has the adapters we want to
#
#       The specific format of this folder is IMPORTANT bey
#   ## seedMismatches: specifies the maximum mismatch count which will
#   ## palindromeClipThreshold: specifies how accurate the match betwee
#   ## simpleClipThreshold: specifies how accurate the match between an
#   ## minAdapterLengthPalindrome: (optional int) specifies the minimum
#   ## keepBothReads: (optional boolean) specifies if both reads should
#
#
#   MAXINFO uses their special algorithm to maintain maximum read length
#
#       Its clever and should do good for us
# Remove leading low quality or N bases (below quality 3) (LEADING:3)
# Remove trailing low quality or N bases (below quality 3) (TRAILING:3)
# Scan the read with a 4-base wide sliding window, cutting when tethe a
#
# Drop reads below 36 bases long (MINLEN:36)
```

- **FastQC of 5927 redo trimmo:** 01/30/26 Alas the previous trimmo still

didn't have the right adapters. Rerunning with a different adapter library. Really wish I had the original information on which adapter Illumina used.

- **Trimmo of 5927 V3:** 01/30/26 Same trimmo but with the TruSeq3-PE-2.fa adapter library to trim.

```
java -jar "./Trimmomatic/target/trimmomatic-0.40.jar" PE -threads $number_t
-trimlog "$trim_log_file" -summary "$summary_file" -compressLevel 9 -version
"$full_input_file_path_1" "$full_input_file_path_2" \
"$full_output_file_path_1_paired" "$full_output_file_path_1_unpaired" \
"$full_output_file_path_2_paired" "$full_output_file_path_2_unpaired" \
HEADCROP:15 \
ILLUMINACLIP:Trimmomatic/adapters/TruSeq3-PE-2.fa \
LEADING:3 \
TRAILING:3 \
MAXINFO:150:5 \
MINLEN:36
```

- **FASTqc of Trimmo 5927 V3 Samples:** 02/02/26 Running fast qc of the trimmo samples I ran last friday. Lets hope it trims the right stuff this time! IT WORKED!!! This is fantastic. Most samples pass all checks with flying colors. Obviously the sample length distribution isn't perfect but that is inherent to the trimming process.

1. JEL272: had significant bimodal per sequence gc content. Really bad. Not sure what to do consider dropping this sample
2. JEL216: had moderate bimodal per sequence gc content. Consider additional cleaning or dropping sample.

- **ALIGNMENT of TRimm0 results** Okay now that trimmo results have been QC'd and all passed well we need to align. I will align using 3 different tools.

1. Bowtie Bowtie was really quite fast and seemed to provide good results. One sample aligned very poorly though. I think that was corrupted anyway. Settings were default however set to "very sensitive"

```
conda run -n bowtie_env bowtie2 -x "$reference_genome_name"
```

2. BWA-MEMs BWA-Mem was slower than bowtie but initial checks seemed quite acceptable. Need to qc more though! Settings were all default:

```
./bwa-mem2-2.2.1_x64-linux/bwa-mem2 mem -o "$sam_save_file" -t
```

3. Stampy Stampy aligned VERY slowly. About a day for all the paired alignments. That being said, I did not enable the speed up BWA co-alignment feature which should increase speed significantly but may decrease accuracy. I also didn't multithread it much so that also impacted speed. Settings were default but set to very sensitive.

```
conda run -n $conda_envron ./stampy/stampy.py -g near_complete
```

- **SeqyClean:** (02/09/26 ish) used seqy clean to clean all the samples again. Testing against Trimmomatic. Notably slower however upon initial assessment it seems to have trimmed less? Not sure if that is better or worse... Need to check quality with fastqc still and push through pipeline. Regardless it seems okay! Most settings were default with specifics here:

```
./seqyclean/bin/seqyclean -qual -dup -detrep -at 0.75 -v "$vector_file" -1
```

- **UPDATE:** (04/01/26) It has been a little bit since we looked at what is going on. So let's update and describe what we have going on right now! I have cleaned all of Imani's samples with both SeqyClean and Trimmomatic.
 - All of Imani's Samples have been cleaned with both Trimmomatic and Seqy clean
 - Trimmomatic samples have been checked with FastQC
 - Trimmomatic Samples have been aligned with BWA-Alignment, Stampy, and Bowtie
 - SeqyClean samples still need fastqc and alignments.
 - Will re-run BWA and Bowtie alignments of Trimmomatic Files as well.
- **bwamem:** (04/01/26) Re-running bwa-mem on Imani Trimmomatic Samples. This is because I can't find the original samples and want to make sure I have good alignments. Doing it in a single run even though it is slower. Will make the files a little easier to track in the end I think.
- **bwamem:** (04/01/26) Re-running the bowtie alignment of Imani Trimmomatic Samples. This is because I can't find the original samples and want to make sure I have good alignments. Doing it in a single run even though it is slower. Will make the files a little easier to track and in the end I think this will help.
- **Fastqc:** Ran all Imani seqy clean samples on fastqc with same settings as above (04/01/26)
- **bwamem:** Ran all Imani seqy clean samples on BWA with same settings as above (04/01/26)

- **stampy:** Ran all imani seqy clean samples on Stampy with same settings as above (04/01/26)
- **NOTE:** (04/04/26) All recent bowtie runs appear to not be saving. I think it needs to be saved in a folder called "sam output file" or something like that. However that folder doesn't exist thus it isn't saved. No matter for now I think I will stick with bwa mems and Stampy.
- **Read groups:** (04/06/26-04/08/26) It appears that both stampy and BWA either drop or simply ignore readgroups. This is not an issue for me as I only have one read/library per sample. However if I was to merge multiple libraries post sequencing to increase my read accuracy then I would want to have read groups to identify which library was run when etc. This isn't much of an issue for me however Picard won't run properly when the read groups identifier is invalid (as I have checked with the sam file validator). Thus I will go through and modify all sam files and add a read group ID to a specific made up string. NOTE: Took a few days for it to all run. This is because the pod didn't have Java on all the nodes so it was delayed randomly. Didn't do anything else during this period though so it should be clear.
- **Coordinate Sorted:** 04/11/2026 Neither BWA nor STAMPY coordinate sort the samples. For Picard to work we MUST coordinate sort all of our sequences. SOooooooooo lets do that with Picard Sort Sam function.
- **Coordinate Sorted:** 04/13/26 Coordinate sort with Picard Sort Sam function on the ss samples. The initial run was done on the test.txt file not the true samples.
- **Mark Duplicates Picard:** 04-13-2026 Used picard to mark duplicates on SS, ST, BS, BT
- **Download and check:** But first let's download all these samples onto the laptop and look at them in geneious
- **Move samples:** 05-29-2026 for longer term storage. Moved the BWAMems, seqy-clean outputs, picard-mark-duplicates, stampy. Need to move more but it takes a bit and requires so many login nodes. Probably could automate this tbh moved
- **UPDATE:**(09/04/26)

Again it has been a little while since I worked with these samples. I should

I will do this analysis specifically on:

— /home/cmallory/Genetic_analysis/bd_genetics_analysis/picard/sort_sa

- `/home/cmallory/Genetic_analysis/bd_genetics_analysis/picard/sort_sa`

The plan now will roughly follow these two methods:

- Allele frequency of heterozygous positions
- Read Depth Coverage (RDC)
- **Method Outline: Allele Frequency of heterozygous positions** This method will roughly follow these steps:
 - Perform initially clean and alignment - Completed already
 - Mark Duplicates (picard) - Completed already
 - Realign with GATK RealignerTargetCreator - NO LONGER NEEDED IN GATK4
 - Obtain SNP records with GATK HaplotypeCaller (minimal at 30, minomum 10phred)
 - GATK SelectVariants - To identify only SNPs and not insertaions and deletions
 - VCFfilter for SNPS with read depth ≥ 10 and quality higher than 10phred
 - Retrieve all HETEROZYGOSE SNPS
 - Filter heterozygous SNPS for only SNPs where all variants had at least a read depth of 5
 - Calculate the proportion of read depth of variant allele (round to the nearest 0.1)
 - Plot distribution of base/variatsn frequency across the chromosome
- **Method Outline: Read Depth Coverage (RDC)**

Theoretically the read depth of a chromosome should be proportional to the Chromosomes with elevated copy numbers should have a deeper read depth than In most systems where people look for an abberhant copy number (meaning tha

- Calculate the MEDIAN of whole genome coverage (excludng Mitochondrial genome and other regions of variable GENE copy number). They excluded known Expression Site Associated Genes (ESAG) and Variant Surface Glycoproteins (VSGs). I will exclude the mitochondrial genome for sure, but I think I will not filter out ESAG and VSGs since I don't know where those are in Bd.
- Calculate the MEDIAN coverage of each chromosome following the same process as above.
- Exclude any genes with coverage $\leq 50\%$ of the length of the gene.

- Exclude any genes with outlier coverage based on the iterative Grubbs Test with $p=0.05$
- Calculate dc/dg for each chromosome
- **NOTE on DetermineGermlineContigPloidy:** GAKT4 has an interesting tool called DetermineGermlineContigPloidy. This tool uses read depth to estimate the total ploidy of each contig. This does not work well in mosaic populations (where there may be subpopulations with different copy numbers). Mosaic populations would yield a fractional copy number estimate which is rounded and distorted by the software. If we assume that my cells are indeed clonal and that there wasn't variance within my sequenced population of cells this tool should work fine. Largely this would replace the RDC estimate noted above. This should be complimented by a SNP variant assessment as well. This tool only uses the read depth to estimate contigs.

Additional note. We must create a good prior estimate for the model however. By default this tool does not allow more than trisomy. This should be updatable by forming and creating an additional TSV with the proper priors.

Note on priors: With good read depth the priors really shouldn't matter too much. This is good because I do not have a great way to estimate the priors. Perhaps I could use the data that Allie had back in their old paper but I am not sure. Fundamentally the diploid state is most probable for all samples but I do not know how to estimate any of the others. That being said from what I understand with good read data the priors really do not matter much.

To use this tool we must follow this protocol

- GAKT4 PreprocessIntervals - This should be performed on the reference sequence!
- GAKT4 CollectReadCounts - This should be performed on the sample
- GAKT4 DetermineGermlineContigPloidy - This should be performed on the sample
- **Pac bio sequence batch start:** 05-29-2026 Hi there world
- **Pac bio sequence batch start:** 06-01-26 I just got my first batch of pac bio sequences back from UC davis. There appears to be a significant coverage of the whole genome which is rad. I don't have any Hi-C sequencing but I think this will be just fine. Conveniently these samples appear to be cleaned prepped and converted to fastq and fasta files directly from UC davis so they should be ready for assembly. I am conflicted with how I want to do the assembly. I think I want to do a denovo assembly first to see how they look and if they look good then keep on doing that for

all. Otherwise I will assemble with the nice pacbio reference genome as I did with all of Imani's NGS samples. My first stab at the assembly will be with the "hifiasm" software. I don't think this is ideal software since it doesn't appear to accommodate polyploids super well... but I am not sure. It also tries to collapse everything into a single contig for inbred/homologous genomes which might be appropriate for us... but that being said we might have multiple alleles so I think at first at least I will attempt the assembly with just the regular heterozygote genome assembly settings and then redo with the inbred/homologous genomes. I will also eventually explore other options for assembly.

- **hifiasm homologous genome** 06-01-26 Started the denovo assembly of the homozygous genome assembly with hifiasm. following these specific parameters:

The homo run:

```
./hifiasm/hifiasm -o $save_file -t$number_of_threads --hg-size 24m -l0 $fastq_file
```

The hetero run settings

```
./hifiasm/hifiasm -o $save_file -t$number_of_threads --hg-size 24m --primary -l1 --n
```

- **hifi-asm** Okay scratch the hifi asm stuff here. I will make another section where I will be tracking the hifi assembly and processing work!
- **Statistics Imani samples** 06-22-24 and 06-23-24 I wanted some statistics on how the assembly went! So using samtools I ran:

1. Flagstats = which should give statistics on the whole assembly prior to generating the consensus sequence

```
conda run -n $conda_environ samtools flagstats $sam_file_path > $sam_res
```

2. idxstats = Which should give contig specific coverage information

```
conda run -n $conda_environ samtools idxstats $sam_file_path > $sam_res
```

3. stats = Which should give statistics on the sequencing results themselves but doesn't particularly comment on the assembly

```
conda run -n $conda_environ samtools stats $sam_file_path > $sam_result
```

These all ran with no particular setting changes. I then ported all the results over to the lenovo so I can assess them later. Upon a brief look through STAMPY and SeqClean appear to perform slightly better than BWA-MEM and Trimmomatic. That being said they are all pretty darn close so idk. Probably doesn't really matter that much!

- **Consensus Sequence** 06-24-26 Imani samples. I ran the Sam tools consensus sequence tool on all the bt, bs, ss, and st samples. Using standard settings (see below). Saving the results into a fastq file. It is using the samtools base Bayesian consensus tool. I wanted it to keep any insertions but it will not replace sections with the reference genomes sequence. So we ultimately should only have the sequenced genome (I think).

```
conda run -n $conda_envIRON samtools consensus -f fastq $sam_file_path -@ $num_
```

8 PacBio_{sequences})

Batch 1 refers to the first batch of 6 samples I had run on the pacbio revio at UCDavis. This first batch of samples included EX124, EX178, EX179, EX187B, EX189B, and EX191A. If I only do something to ONE sequence then I will call out that sequence alone. Otherwise if I say "Batch 1" that will reference this entire group of samples.

- Batch 1: EX124, EX178, EX179, EX187B, EX189B, EX191A

Here begins the notes of what was done when for the pac bio sequences:

- **PAC BIO SEQUENCES BATCH 1 START** 06-01-26 I just got my first batch of pac bio sequences back from UC davis. There appears to be a significant coverage of the whole genome which is rad. I don't have any Hi-C sequencing but I think this will be just fine. Conveniently these samples appear to be cleaned prepped and converted to fastq and fasta files directly from UC davis so they should be ready for assembly. I am conflicted with how I want to do the assembly. I think I want to do a denovo assembly first to see how they look and if they look good then keep on doing that for all. Otherwise I will assemble with the nice pacbio reference genome as I did with all of Imanis NGS samples. My first stab at the assembly will be with the "hifiasm" software. I don't think this is ideal software since it doesn't appear to accommodate polyploids super well... but I am not sure. It also tries to collapse everything into a single contig for inbred/homologous genomes which might be appropriate for us... but that being said we might have multiple alleles so I think at first at least I will attempt the assembly with just the regular heterozygote genome assembly settings and then redo with the inbred/homologous genomes. I will also eventually explore other options for assembly.

- **hifiasm** In the end I do not think hifiasm is the way to go. It doesn't resolve haplotypes super well by itself. It needs HiC data which is fine I guess. That being said it can create partially phased contigs! However you need to be sure not to include the -primary or the -l0 flags otherwise it will create only the primary contigs. I don't want the code to deduplicate things at all at this stage. So if I use the -l0 phasing that should prevent the duplication purging. I will later need to sort stuff out based on read depth but I can do that with other post processing software I think. Additionally I think I should probably just use a different denovo alignment software. But we will get to that in time!

```
./hifiasm/hifiasm -o save_file -t number_of_threads -hg -size24m -l0fastq_file
```

- **hifiasm build the assembly** 06-23-26 I still don't think hifiasm is the final way to go. That being said I do want to convert it to an assembly that I can use and look at! So let's figure out how to do that and import it over to Genious on

my laptop to better understand what it puts out! FOr starters I want to focus on EX124 from Batch 1. Currently it has several file types that are importat (yes they are ALL important but a few are most intersting)

- ‘prefix’.*r_utg.gfa* : *haplotype-resolvedrawunitiggraph.Thisgraphkeepsallhaplotypeinformation.* ‘p
haplotype – resolvedprocessedunitiggraphwithoutsmallbubbles.
- ‘prefix’.*p_ctg.gfa* : *assemblygraphofprimarycontigs.* ‘prefix’.*a_ctg.gfa* : *assemblygraphofalternatcontigs.*
 The software can create more file types than these. But with the settings I used for my runs (essnetially only the homo run method) it should only creat these files. In the end I want a different tool but this is what I am using now.
- **GFAVIZ** 06-23-26 I have also installed gfaviz onto my personal laptop to try and see if I can visualize the gfa files. So far tthe tool has been awful to get working.
- **Note:** TEXShop will not save your latex file if it cannot full run everything (e.g. if there is an error). I just lost a whole days worth of notes and comments on alternate methods for assembly and a whole discussion on what I think is best in this context. BE cautious and just use VS code cause it will save no matter what